OpportunIQ

AI-Powered Resume, Cover Letter & Career Gap Intelligence Platform

Team ID: H26EDU11

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Project Details

- **Project Title:** OpportunIQ
- **Thrust Area:** AI for Education & Career Readiness
- **Hackathon:** TATA Centre AI/ML Hackathon 2026
- **Venue:** NIT Tiruchirappalli

Team Details

- **Team Lead:** Anantha Ram G S
- **Team Member:** Gowri J S
- **Team Member:** Sanjay Anand M

Problem Statement & Objectives

Problem: Manual Resume Tailoring is Inefficient

- Students apply to many roles, but each resume must be customized for a specific job description.
- Projects, skills, education, certifications and experience are scattered across documents and manual notes.
- Most resumes are not optimized for ATS keywords, role relevance or measurable impact.
- Students lack clear guidance on what skills or projects are missing for their target career path.

Target Users

- University students and fresh graduates.
- Placement-ready candidates applying to internships and entry-level roles.
- Career services teams that need scalable resume support.

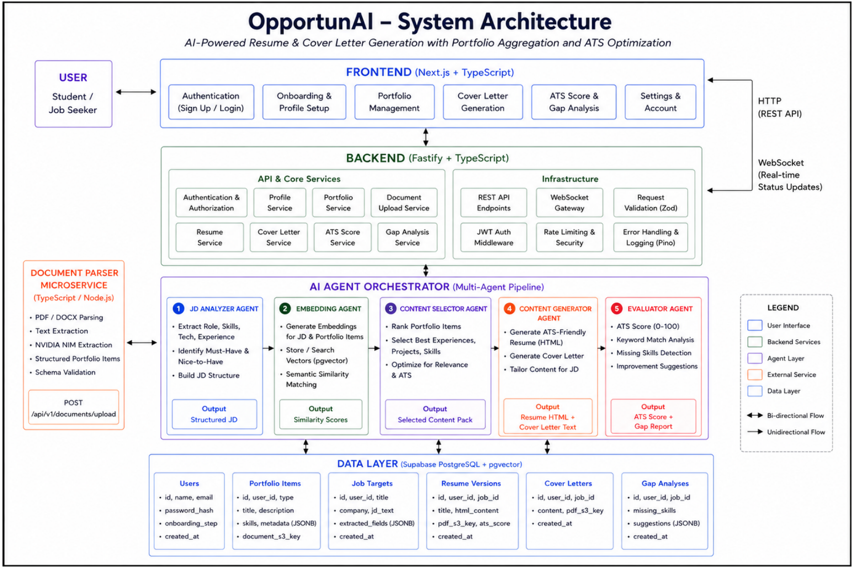
Objectives of OpportunIQ

1. Build a unified portfolio workspace for projects, education, skills, certifications and experience.
2. Generate ATS-friendly resumes automatically from a target job description.
3. Generate role-specific cover letters linked to the resume version.
4. Provide ATS scoring and career gap analysis to guide improvement.

Core Value Proposition

- From **portfolio + job description** to **tailored PDF resume** in seconds.
- Converts static student data into an AI-powered job application assistant.

System Design



Algorithms

1. Document-to-Portfolio Extraction

- PDF/DOCX is uploaded to AWS S3 and parsed using `pdf-parse` or `mammoth`.
- Extracted text is sent to NVIDIA NIM for structured portfolio extraction.
- AI output is validated using Zod portfolio schemas before database persistence.
- Invalid extracted items are discarded; fully invalid extraction fails safely.

2. JD Analysis & Portfolio Scoring

- Extract required skills, preferred skills, role category and seniority from the job description.
- Score each portfolio item using keyword relevance, impact metrics, recency and type weight.
- Select the most relevant items within resume page constraints.
- Preserve user ownership by filtering all portfolio data by authenticated user ID.

3. Resume & Cover Letter Generation

- Selected portfolio items and JD entities are converted into structured resume content.
- Resume HTML templates are rendered into PDF using Puppeteer.
- Generated PDFs are uploaded to S3 and accessed through signed URLs.
- Cover letters reuse resume context, JD context and user-provided tone/motivation.

4. ATS Scoring & Gap Advisor

- ATS score compares generated resume content against extracted JD keywords.
- Found and missing keywords are returned with improvement suggestions.
- Gap Advisor compares career goal with current portfolio signals.
- Produces missing skills, suggested projects and learning recommendations.

Technology Stack

Frontend

- React + Vite
- TypeScript
- React Router
- Axios API client
- TipTap resume editor

Backend

- Node.js + TypeScript
- Fastify
- Zod validation
- JWT authentication
- Puppeteer PDF rendering

AI / ML

- NVIDIA NIM
- Google Gemini
- Groq client retained for AI modules/fallbacks
- Deterministic ATS scorer
- Prompt-based structured JSON extraction

Database

- PostgreSQL via Supabase
- Raw SQL migrations
- UUID primary keys
- JSONB AI outputs
- User-scoped relational model

Storage & Infra

- AWS S3 private bucket
- Signed URLs
- Docker PostgreSQL for local dev
- Supabase pooler for cloud DB
- pnpm + Vitest + ESLint

Testing & Validation

- 90 backend tests passing
- Unit + route tests
- S3 smoke test
- Cloud DB smoke flow
- Type-check, lint, build gates

Results & Demo

Key Results

- End-to-end backend implemented for Auth, Profile, Portfolio, Upload, Settings and Resume workflows.
- Supabase PostgreSQL schema migrated and verified in cloud.
- AWS S3 upload, signed URL generation and delete flow validated.
- Document upload supports PDF/DOCX extraction into portfolio items.
- Resume generation creates versioned PDF resumes with ATS feedback.
- Cover letter and gap analysis workflows are integrated into the application.

Demo Video

Click Here to View Demo Video

Comparison with Existing Solutions

Existing Solutions

- **Resume Builders:** Provide templates but require users to manually rewrite content for every role.
- **ATS Checkers:** Score resumes but do not generate improved, role-specific versions automatically.
- **Generic LLM Chatbots:** Generate text but lack portfolio memory, versioning and database-backed workflows.
- **Job Boards:** Provide listings but do not transform student portfolio data into optimized applications.
- **Cover Letter Tools:** Generate letters in isolation without resume/JD context.
- **Career Guidance Platforms:** Provide advice but rarely connect it to resume generation and portfolio assets.

Identified Gaps

- × No unified student portfolio asset pool.
- × No automatic selection of the most relevant projects and experience per JD.
- × No combined resume generation, ATS feedback and cover letter flow.
- × No stored resume version vault for repeated applications.
- × No document upload pipeline that converts old resumes into structured portfolio items.
- × No career gap analysis linked to the user's actual portfolio.

OpportunIQ's Unique Contribution

OpportunIQ combines **portfolio intelligence** + **JD-aware resume generation** + **ATS scoring** + **cover letters** + **gap analysis** in one end-to-end career application workflow.

Summary & Future Work

Summary

- Built an AI-powered resume generation platform for students and fresh graduates.
- Unified manual entries and document uploads into a structured portfolio database.
- Generated tailored resume PDFs using job description analysis and portfolio scoring.
- Added cover-letter generation, ATS feedback and career gap analysis.
- Integrated Supabase PostgreSQL and AWS S3 for cloud-ready data and file storage.
- Hardened upload flow with schema validation, route tests and S3 cleanup on failure.

Future Work

- Add GitHub sync for project import and enrichment.
- Add browser extension to capture job descriptions from job portals.
- Add queue-based async processing and WebSocket progress streaming.
- Add vector embeddings and semantic portfolio retrieval.
- Add frontend automated tests and full E2E test coverage.
- Add monitoring, error tracking, CI/CD and production deployment runbooks.

How Can This Become a Product?

Target Users & Business Model

- **Primary Users:** University students, fresh graduates and early-career professionals.
- **Institutional Users:** Placement cells and career development departments.
- **Freemium Model:** Free resume builder with premium AI generation, ATS insights and cover letters.
- **Revenue:** Student subscriptions, institutional licensing and placement-cell dashboards.

Growth Roadmap

- **Phase 1:** Pilot with students for resume tailoring and portfolio upload.
- **Phase 2:** Add GitHub sync, extension and semantic scoring.
- **Phase 3:** Institution-grade platform for placement teams.

Scaling for Production

- Queue-based AI jobs for stable high-volume resume generation.
- Strict IAM, S3 lifecycle policies and account-level asset cleanup.
- CI/CD, monitoring, error tracking and audit logs.
- Multi-tenant design for colleges and placement departments.

Future Product Features

- AI career coach for interview preparation and learning plans.
- Browser extension for one-click JD import.
- GitHub and LinkedIn enrichment.
- Campus placement dashboard and analytics.